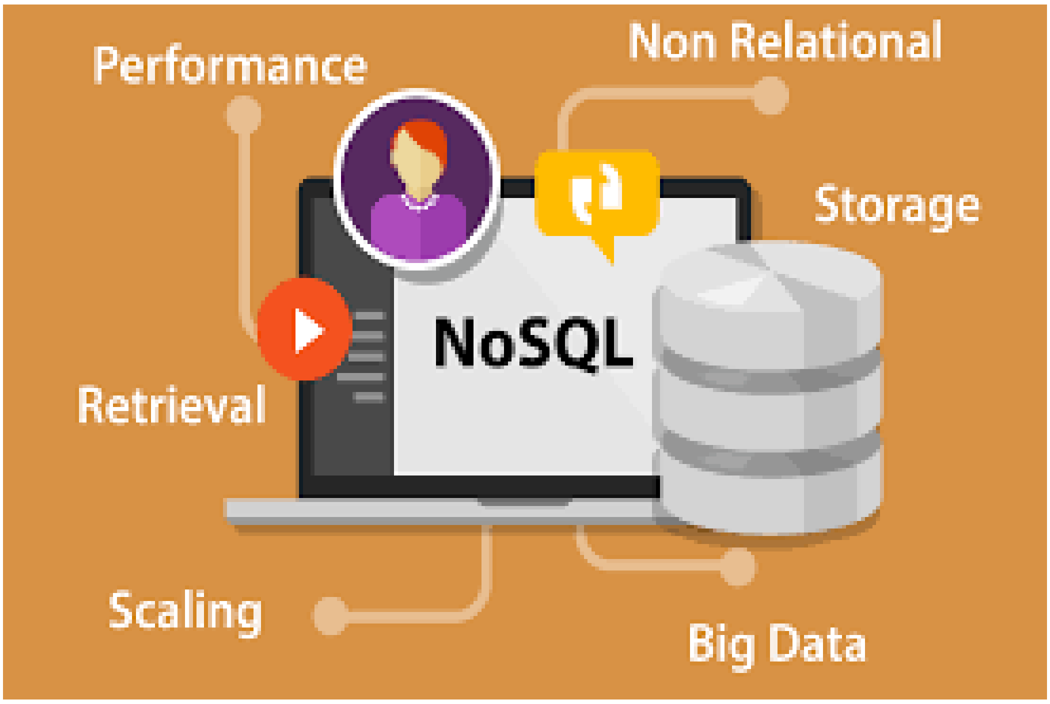
Mongo DB & NoSQL DB

Presnted By,

J. Steve Manuel

NoSQL DB



NoSQL ("Not Only SQL") refers to a broad category of databases that don't use the traditional table-based relational model.

- **Dynamic Schema (Flexibility):** Unlike SQL, NoSQL allows you to insert data without a predefined schema. This is perfect for "Big Data" or applications where requirements change frequently.

- **Horizontal Scalability (Scaling Out):** Instead of buying a bigger, more expensive server (Vertical Scaling), NoSQL databases scale by adding more cheap servers to a cluster. This makes it much more cost-effective to handle millions of users.

- **High Performance:** Because many NoSQL databases avoid complex "JOIN" operations—which are computationally expensive—they can read and write data much faster than SQL at high volumes.

- **Availability:** Most NoSQL systems are designed to be "distributed," meaning they replicate data across multiple locations. If one server goes down, the database stays online.

MongoDB (The Document Model)

MongoDB is the most popular type of NoSQL database. It is a **Document Store**, meaning it stores data in a format called **BSON** (Binary JSON).

Key Advantages:

- **Matches Modern Code:** Developers usually work with "Objects" in languages like JavaScript (JSON), Python (Dictionaries), or Java. MongoDB stores data in that exact same format. This removes the need for complex "Object-Relational Mapping" (ORM) layers.

- **The "One Document" Advantage:** In SQL, a single "User Profile" might be split across 5 different tables (Users, Addresses, Settings, etc.), requiring a slow JOIN to fetch. In MongoDB, you store all that related info in **one document**. Fetching it takes a single, lightning-fast request.

- **Native Sharding:** MongoDB has a built-in feature called **Sharding**. It automatically breaks your data into chunks and moves them to different servers as your database grows, ensuring no single server gets overwhelmed.

- **Rich Querying & Aggregation:** Unlike some NoSQL databases that only let you search by ID, MongoDB has a powerful "Aggregation Pipeline" that allows you to perform complex data analysis (grouping, filtering, averaging) directly within the database.

REAL TIME USE CASES OF NoSQL DB

➢ Real-Time Personalization & Recommendation Engines

Streaming giants and e-commerce sites use NoSQL to track user behavior (clicks, watch time, searches) in real time. Because the data is "unstructured" (different users interact with different items), a flexible schema is required to update user profiles instantly and serve "Because you watched..." suggestions.

- **Why NoSQL:** Low latency for "reads" ensures recommendations appear instantly without lagging the UI.

➢ Internet of Things (IoT) & Sensor Data

IoT networks involve thousands of sensors sending a constant stream of data (temperature, GPS, pressure) every second. This data is often "time-series" and can vary in format between different device models.

- **Why NoSQL:** Databases like MongoDB or Cassandra can handle the high "write" volume of millions of small data packets per second and scale horizontally as more devices are added to the network.

➢ Gaming Leaderboards & User Profiles

Modern multiplayer games need to store massive amounts of player data, including inventory, achievements, and real-time rank. When millions of players are online, updating a global leaderboard requires a database that can handle rapid, concurrent updates.

- **Why NoSQL:** It allows for fast updates to complex "player objects" (like an inventory with 100 items) without needing to perform slow SQL JOINs across multiple tables.

➢ Content Management Systems (CMS) & Personalization

Platforms like Forbes or eBay use NoSQL to manage diverse content types (articles, videos, comments, metadata). Since a "Video" post has different fields than a "Text" post, a rigid table structure is inefficient.

- **Why NoSQL:** The **Document Model** allows you to store all related content for a single page in one place, making it incredibly fast to serve the entire page to a user.

➢ Fraud Detection & Real-Time Analytics

Financial institutions monitor transactions as they happen to flag suspicious activity. This requires comparing a current transaction against a user's historical patterns and global fraud signatures in milliseconds.

- **Why NoSQL:** NoSQL databases (especially Graph databases or Key-Value stores) excel at finding relationships between data points quickly enough to block a fraudulent transaction before it is even processed.